numerical terms, it is equivalent to approximately one millionth of a billionth of a billionth of a billionth of a centimetre (10^{-33} centimetres). So, that is to say … small.

It is at this extremely minute scale that quantum gravity manifests itself. To give an idea of the smallness of the scale we are discussing: if we enlarged a walnut shell until it had become as big as the whole observable universe, we would still not see the Planck length. Even after having been enormously magnified thus, it would still be a million times smaller than the actual walnut shell was before magnification. At this scale, space and time change their nature. They become something different; they become 'quantum space and time', and understanding what this means is the problem.

Matvei Bronštejn understands all of this in the 1930s and writes two short and illuminating articles in which he points out that quantum mechanics and general relativity, taken together, are incompatible with our customary idea of space as an infinitely divisible continuum.[2]

There is, however, a problem. Matvei and Lev are sincere communists. They believe in revolution as the liberation of mankind, the construction of a genuinely better society, without injustice, without the immense inequalities which we still see growing systematically throughout the world. They are enthusiastic followers of Lenin. When Stalin assumes power, they are both perplexed, then critical, then hostile. They write articles which are mildly but openly critical … This was not the communism they wanted …

But these are harsh times. Landau gets through them, not easily, but he survives. Matvei, the year after having been the first to understand that our ideas on space and time had to change in a radical way, is arrested by Stalin's police and condemned to death. His execution takes place on the same day as his trial, 18 February, 1938.[3] He is thirty years old.

John

After Matvei Bronštejn's premature death, many of the century's eminent physicists tried to solve the puzzle of quantum gravity. Dirac dedicated the final years of his life to the problem, opening avenues and introducing ideas and techniques on which a good part of current work on quantum gravity is based. It is thanks to these techniques that we know how to describe a world without time, as I will explain further on. Feynman tried, attempting to adapt the techniques he had developed for electrons and photons to the context of general relativity, but without success: electrons and photons are quanta in space; quantum gravity is something else: it isn't enough to describe 'gravitons' moving in space, it is space itself that has to be quantized.